

THE DENSITY OF GRAPHS WITH NO ℓ -PATH CONNECTING EQUAL-DEGREE VERTICES: A SHORT PROOF

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ABSTRACT. Addressing a question posed by Chen and Ma from an asymptotic point of view, we present a short proof for the edge density needed to guarantee that two vertices of the same degree are connected by a path of a fixed length. In particular, we show that for any sufficiently large graph, a density of at least $1/2 + o(1)$ enforces the existence of two such vertices. This bound is tight for paths of odd length.

1. INTRODUCTION

It is a simple exercise to show that every graph contains a pair of vertices of the same degree. Being so, it is only natural to ask: under what conditions can we expect such vertices to satisfy additional properties? Erdős and Hajnal [2] formulated the following question of this flavor:

Problem 1.1. Is it true that every $(2n + 1)$ -vertex graph with $n^2 + n + 1$ edges contains two vertices of the same degree which are joined by a path of length three?

As usual, the length of a path is the number of edges it contains.

The bound of $n^2 + n + 1$ is tight, as demonstrated by the complete bipartite graph $K_{n,n+1}$. In a recent paper, Chen and Ma [1] resolved this problem, in a very strong sense, for large n . In fact, they also handled the case where the graph has an even number of vertices, characterizing the unique extremal construction in both situations.

Theorem 1.2. *The following holds.*

- (a) *Let $n \geq 600$. The unique $(2n + 1)$ -vertex graph with at least $n^2 + n$ edges that does not contain two vertices of the same degree joined by a path of length three is the complete bipartite graph $K_{n,n+1}$.*
- (b) *There exists an integer $n_0 > 0$ such that the following holds for all $n \geq n_0$. The unique $2n$ -vertex graph with at least $n^2 - 1$ edges that does not contain two vertices of the same degree joined by a path of length three is the complete bipartite graph $K_{n-1,n+1}$.*

This result was later improved by Liu and Zeng [3], who showed that part (a) holds for all $n \geq 2$ and part (b) is true for all $n \geq 3$.

In their paper, Chen and Ma [1] defined the following extremal function, generalizing the Erdős–Hajnal problem.

Definition 1. For any positive integers ℓ and n , let $p_\ell(n)$ denote the maximum number of edges in an n -vertex graph G that contains no two vertices of equal degree connected by a path of length ℓ .

The results in [1, 3] thus show that $p_3(2n + 1) = n^2 + n$ for all $n \geq 2$ and $p_3(2n) = n^2 - 1$ for all $n \geq 3$. Chen and Ma further showed that

$$p_1(n) = \frac{n^2}{2} - \frac{n\sqrt{2n}}{3} + O(n) \quad \text{and} \quad p_2(2n) = \frac{n(n+1)}{2}.$$

Additionally, they conjectured that the behavior of $p_\ell(2n + 1)$ should be the same for any odd $\ell \geq 3$.

Conjecture 1.3. *For any odd integer $\ell \geq 3$ and sufficiently large n , it holds that $p_\ell(2n + 1) = n^2 + n$.*

In a very recent paper, Liu and Zeng [4] tackled the case $\ell = 5$, once again in a strong sense, proving an analog of both parts of Theorem 1.2 (for $n \geq 11$ in part (a) and $n \geq 13$ in (b)). The conjecture was subsequently resolved in full by Zhao, Wang, and Lu [5], who once again proved that $K_{n,n+1}$ is the unique extremal example when the number of vertices is odd. The case of an even number of

vertices was not addressed directly in [5], even though the authors comment that their methods likely extend to this setting.

For even values of ℓ , our knowledge is much more limited. The *half graph* on $2n$ vertices is defined as the bipartite graph with vertex classes $\{u_1, \dots, u_n\}$ and $\{v_1, \dots, v_n\}$, where $u_i v_j$ is an edge if and only if $i \leq j$. It is not difficult to check that in the half graph no vertices of the same degree are connected by an even-length path, implying that $p_\ell(2n) \geq \frac{n(n+1)}{2}$ for all even $\ell \geq 2$. As far as we are aware, nothing more is known about this function when ℓ is even, though in [4] Liu and Zeng refer to upcoming work showing that $p_4(2n) \leq (1 - \varepsilon + o(1))n^2$ for some constant $\varepsilon \in (0, 1/2]$ and sufficiently large n .

In all previous work on the problem, the arguments have been rather technical and involved. The goal of this note is to provide a short proof of an asymptotic version of Conjecture 1.3. Our main result addresses the remaining cases by showing that the number of edges in a graph not containing a pair of vertices connected by a path of some fixed length $\ell \geq 6$, odd or even, is always at most $\frac{1}{2} \binom{n}{2}$ asymptotically. Our analysis is somewhat reminiscent of that provided in [5] but it is much simpler.

Theorem 1.4. *For every $\ell \geq 6$ and every sufficiently large n , we have $p_\ell(n) \leq (\frac{1}{4} + o(1))n^2$.*

Clearly, this bound is tight when ℓ is odd. We tend to believe that it should not be tight for even ℓ and in that case the maximum number of edges should be closer to $\frac{1}{4} \binom{n}{2}$. We pose this as a question.

Question 1.5. Is it true that $p_\ell(n) = (\frac{1}{8} + o(1))n^2$ for all even ℓ ?

Notation. Our notation is mostly standard. An xy -path is a path connecting vertices x and y . Let G be a graph and $A, B \subseteq V(G)$ be any subsets (not necessarily disjoint). We write $G[A]$ for the subgraph of G induced by A , that is, the subgraph obtained by deleting the vertices not in A . The neighborhood of A , denoted $N(A)$, refers to the set $\{v \in V(G) : va \in E(G) \text{ for some } a \in A\}$. We write $E(A, B)$ for the set $\{ab \in E(G) : a \in A, b \in B\}$.

2. THE PROOF

This section is devoted to the proof of Theorem 1.4. Throughout this section, we will always assume that $\ell \geq 2$ is a fixed integer, n is sufficiently large with respect to ℓ , and G is an n -vertex graph. The proof of Theorem 1.4 relies on the following technical lemma, which allows us to find long paths when G has too many vertices of large enough degree. We consider several cases depending on whether we look for paths of even or odd length.

Lemma 2.1. *Let $D \geq \frac{n}{2} + k + 1$ be an integer. Suppose G contains a set $B = \{b_1, \dots, b_t\}$ of $2 \leq t \leq k$ vertices of degree at least D , and let $x, y \notin B$ be distinct vertices of degree at least $n - D + t + 4$. Then:*

- (a) *There exists an xy -path of length $2t + 2$.*
- (b) *If $G[B]$ contains an edge, then there exists an xy -path of length $2t + 1$.*
- (c) *If $E(N(b_i), N(b_j)) \neq \emptyset$ for some $i \neq j$, then there exists an xy -path of length $2t + 3$.*

Proof.

- (a) Set $b_0 = x$ and $b_{t+1} = y$. We iteratively find a family of pairwise distinct vertices $a_1, \dots, a_{t+1} \notin B \cup \{x, y\}$ such that a_i is a common neighbor of b_{i-1} and b_i for each $1 \leq i \leq t$. Once this is done, $xa_1b_1a_2 \dots a_t b_t a_{t+1}y$ will form the desired xy -path of length $2t + 2$.

We first choose a_1 and a_{t+1} . Observe that $\deg(b_i) + \deg(b_{i+1}) \geq n + t + 4$ for $i \in \{0, t\}$. Hence $|N(b_i) \cap N(b_{i+1})| \geq t + 4$, and thus this set contains two distinct vertices not in $B \cup \{x, y\}$.

Now, for $1 < i \leq t - 1$, if $a_1, \dots, a_i$ are already selected, we similarly find that $|N(b_i) \cap N(b_{i+1})| \geq 2D - n \geq 2t + 2$. So we can choose the next vertex $a_{i+1} \in N(b_i) \cap N(b_{i+1})$ outside of $B \cup \{x, y, a_1, \dots, a_i, a_{t+1}\}$; indeed, since $b_i \notin N(b_i)$ and $b_{i+1} \notin N(b_{i+1})$, the latter set contains at most $2t$ vertices of $N(b_i) \cap N(b_{i+1})$.

- (b) Assume without loss of generality that $b_1 b_2$ is an edge. We choose $a_1, \dots, a_{t+1}$ as in the previous case, but build a path using the edge $b_1 b_2$ instead of the vertex a_2 .
- (c) Assume without loss of generality that there is an edge $a_2 a'_2$ with $a_2 \in N(b_1)$ and $a'_2 \in N(b_2)$. In this case, we choose $a_1 \in (N(b_1) \cap N(x)) \setminus (B \cup \{x, y, a_2, a'_2\})$, since this set contains at least $t + 4 - (t + 2) \geq 2$ vertices. Similarly, we choose $a_{t+1} \in (N(b_t) \cap N(y)) \setminus (B \cup \{x, y, a_2, a'_2\})$. Since, in the previous items, we had at least two choices for the vertices $a_3, \dots, a_t$, we can also choose them distinct from a'_2 . We thus obtain the path $xa_1 b_1 a_2 a'_2 b_2 a_3 \dots a_t b_t a_{t+1} y$ of length $2t + 3$. $\square$

Using this lemma, we can complete the proof of Theorem 1.4. The proof is split into two cases, depending on the parity of t . We start with the less involved even case.

2.1. Even-length paths. Assume that G does not contain two vertices of the same degree connected by a path of length $\ell = 2k$. Our goal is to show that $e(G) = n^2/4 + O(n)$. Label the vertices of G as $v_1, \dots, v_n$ in non-increasing degree order, and let $\Delta_k = \deg(v_{k-1})$. We consider two cases depending on whether Δ_k is large or not.

Case 1: $\Delta_k < \frac{n}{2} + k + 1$. Summing up the degrees of the vertices, and upper-bounding $\deg(v_i)$ by n if $i < k - 1$ and by Δ_k otherwise, we obtain

$$2e(G) \leq (k-2)n + (n-k+2)\left(\frac{n}{2} + k\right) \leq \frac{n^2}{2} + \frac{3}{2}kn,$$

as claimed.

Case 2: $\Delta_k \geq \frac{n}{2} + k + 1$. Our goal is to apply Lemma 2.1 (a), with $t = k - 1$, $B = \{v_1, \dots, v_{k-1}\}$, and $D = \Delta_k$. If there is at most one vertex $x \notin B$ of degree at least $n - \Delta_k + (k-1) + 4 = n - \Delta_k + k + 3$, we are done by a similar argument as above. So we may assume there exist at least two of them, say x and y . By Lemma 2.1, there exists an xy -path of length $2(k-1) + 2 = 2k$, so we conclude that x and y must have distinct degrees.

Again, summing up the degrees of the vertices, we obtain that $2e(G)$ is bounded above by

$$\begin{aligned} (1) \quad & \underbrace{(k-1)n + \sum_{i=n-\Delta_k+k+3}^{\Delta_k} i}_{\substack{\text{at most one vertex of each degree} \\ v_i \text{ for } i < k}} + \underbrace{(n - (\Delta_k - (n - \Delta_k + k + 3)) + 1)(n - \Delta_k + k + 2)}_{\text{remaining vertices}} \\ &= (k-1)n + \binom{\Delta_k + 1}{2} - \binom{n - \Delta_k + k + 3}{2} + (2n - 2\Delta_k + k + 4)(n - \Delta_k + k + 2) \\ &\leq (k-1)n + \binom{\Delta_k + 1}{2} - \binom{n - \Delta_k + k + 3}{2} + (n - k + 2)(n - \Delta_k + k + 2) \\ &= \frac{\Delta_k^2}{2} - \frac{(n - \Delta_k)^2}{2} + n(n - \Delta_k) + O(n) = \frac{n^2}{2} + O(n), \end{aligned}$$

which concludes the even case.

2.2. Odd-length paths. Assume now that G does not contain two vertices of the same degree connected by a path of length $\ell = 2k + 1$ with $k \geq 3$. We again label the vertices $v_1, \dots, v_n$ by non-increasing degree and set $\Delta_k = \deg(v_k)$.

Let $D = \frac{n}{2} + k + 1$ and $B = \{v \in V(G) : \deg(v) \geq D\}$. Note that $2e(G) \leq n|B| + D(n - |B|)$. If $|B| = o(n)$, then $e(G) \leq \frac{n^2}{4} + o(n^2)$ and we are done. So we may assume that $|B| = \omega(n)$ and in particular that $\Delta_k \geq D$. Additionally, write $R = V(G) \setminus (B \cup N(B))$.

Case 1: $G[B]$ contains an edge uw or $E(N(u), N(w)) \neq \emptyset$ for some distinct $u, w \in B$.

Suppose first that $G[B]$ contains the edge uw . Let x, y be two vertices of degree at least $n - \Delta_k + k + 4$. Select $k - 2$ vertices $b_1, \dots, b_{k-2} \in B \setminus \{u, w, x, y\}$, and write $b_{k-1} = u, b_k = w$. By Lemma 2.1 (b), G must contain an xy -path of length $2k + 1$, and hence x and y have distinct degrees. In particular, no two vertices of B (except possibly u, w) share the same degree. Then proceeding similarly to (1), where we allow for the degrees of u and w to be as large as n , we obtain

$$\begin{aligned} 2e(G) &\leq (k+2)n + \sum_{i=n-\Delta_k+k+4}^{\Delta_k} i + (n - (\Delta_k - (n - \Delta_k + k + 4)) + 1) \cdot (n - \Delta_k + k + 3) \\ &\leq \frac{n^2}{2} + O(n), \end{aligned}$$

using that $n \geq \Delta_k \geq D \geq n/2$.

The latter case, where $E(N(u), N(w)) \neq \emptyset$ for some distinct $u, w \in B$, is similar, except we now take $b_1, \dots, b_{k-3} \in B \setminus \{u, w, x, y\}$, and write $b_{k-2} = u, b_{k-1} = w$ and apply Lemma 2.1 (c).

Case 2: B is an independent set and $E(N(u), N(w)) = \emptyset$ for any distinct $u, w \in B$.

In this case, $B \cap N(B) = \emptyset$. Partition $N(B)$ into sets $X \cup (\bigcup_{v \in B} Y_v)$, where $X = \{w \in V(G) : |N(w) \cap B| \geq 2\}$ and $Y_v = \{w \in V(G) : N(w) \cap B = \{v\}\}$ for all $v \in B$. Note that the only edges within $N(B)$ are fully contained in the sets Y_v . We now claim that X is relatively large.

Claim 2.2. $|X| \geq \frac{|B|-2}{|B|-1} \cdot \frac{n}{2}$.

Proof. We proceed by double counting the edges between B and X . On the one hand, each vertex in X is adjacent to at most $|B|$ vertices in B , so $|E(B, X)| \leq |X| \cdot |B|$. On the other hand, we have

$$\begin{aligned} |E(B, X)| &= |E(B, N(B))| - \sum_{v \in B} |E(B, Y_v)| \geq D|B| - \sum_{v \in B} |Y_v| \\ &\geq \frac{n}{2}|B| - (n - |B| - |X| - |R|) \geq (|B| - 2)\frac{n}{2} + |X|, \end{aligned}$$

where in the first inequality we used that every vertex in Y_v has at most one edge to B . Combining these two bounds proves the claim. $\square$

We now conclude by estimating the number of edges of G . For this, recall that B is an independent set, the only edges within $N(B)$ can occur inside the individual sets Y_v , and every vertex in R has degree at most D by definition. Hence

$$\begin{aligned} e(G) &\leq |B| \cdot |X| + \sum_{v \in B} \left(|Y_v| + \binom{|Y_v|}{2} \right) + D|R| \\ &\leq |B| \cdot |X| + \binom{\sum_v |Y_v|}{2} + \frac{n}{2}(n - |B| - |N(B)|) + O(n) \\ (2) \quad &\leq |B| \cdot |X| + \binom{|N(B)| - |X|}{2} + \frac{n}{2}(n - |B| - |N(B)|) + O(n), \end{aligned}$$

where for the second inequality we used that $\sum_{v \in B} \binom{|Y_v|}{2} \leq \binom{\sum_v |Y_v|}{2}$, and that $R = V(G) \setminus (B \cup N(B))$ and $B \cap N(B) = \emptyset$. Note that, with respect to $|X|$, the expression in (2) is an upward-opening parabola, so its value is maximized at some extreme point for $|X|$. By Claim 2.2 and the fact that $X \subseteq N(B)$, we know that $\frac{(|B|-2)}{(|B|-1)} \cdot \frac{n}{2} \leq |X| \leq |N(B)|$. We consider these two cases separately. Observe that $\frac{n}{2} \leq D \leq |N(B)| \leq n - |B|$, so in particular $|B| \leq n - D < \frac{n}{2}$.

Case 1: $|X| = |N(B)|$. Then (2) becomes

$$\begin{aligned} &|B| \cdot |N(B)| + \frac{n}{2}(n - |B| - |N(B)|) + O(n) \\ &= |N(B)| \left(|B| - \frac{n}{2} \right) + \frac{n^2}{2} - \frac{n}{2}|B| + O(n) \\ &\leq \frac{n}{2} \left(|B| - \frac{n}{2} \right) + \frac{n^2}{2} - \frac{n}{2}|B| + O(n) = \frac{n^2}{4} + O(n), \end{aligned}$$

as required.

Case 2: $|X| = \frac{(|B|-2)}{(|B|-1)} \cdot \frac{n}{2}$. Note that in this case (2) is also an upward-opening parabola with respect to $|N(B)|$. Hence, it is maximized when either $|N(B)| = D$ or $|N(B)| = n - |B| \leq n$. In the latter case, (2) is upper bounded by

$$\begin{aligned} &|B| \cdot |X| + \binom{n - |B| - |X|}{2} + O(n) \leq |B| \cdot |X| + \frac{(n - |B| - |X|)^2}{2} + O(n) \\ &\leq \frac{|B|^2}{(|B|-1)^2} \cdot \frac{n^2}{8} + \frac{|B|^2}{2} + O(n) \leq (1 + o(1)) \frac{n^2}{4}, \end{aligned}$$

since $|B| = \omega(n)$ and $|B| \leq \frac{n}{2}$.

In the case $|N(B)| = D$, we bound (2) by

$$\begin{aligned} &|B| \cdot |X| + \binom{n/2 + k + 1 - |X|}{2} + \frac{n}{2}(n - |B| - |N(B)|) + O(n) \\ &\leq |B| \cdot \frac{n}{2} + \frac{\left(\frac{n}{2(|B|-1)} \right)^2}{2} + \frac{n}{2} \left(\frac{n}{2} - |B| \right) + O(n) = (1 + o(1)) \frac{n^2}{4}, \end{aligned}$$

where we used that $|X| \leq \frac{n}{2}$ and $|B| = \omega(n)$. This concludes the proof for the odd-length case.

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